

GUIDED LESSON · LEARN BY EXPERIMENTING

Build H→Z→H & Follow the Interference

A from-scratch walkthrough for total beginners: combine two gates into one striking result, then break down path by path exactly why it happens.

QUICK REFRESHER — EVERYTHING YOU NEED

AMPLITUDE → PROBABILITY Square an amplitude's magnitude to get a probability. $0.7071^2 \approx 0.5$.	HADAMARD (H) $H 0\rangle = (0\rangle + 1\rangle)/\sqrt{2}$. $H 1\rangle = (0\rangle - 1\rangle)/\sqrt{2}$. Applying H twice returns the original state.
Z GATE Leaves $ 0\rangle$ alone, flips the sign of $ 1\rangle$: $ 1\rangle \rightarrow - 1\rangle$. Probabilities don't change on their own.	CIRCUIT DIAGRAMS A wire = the qubit over time. Boxes = gates, applied left to right. Label before = start, label after = result.

05

Build H → Z → H

Putting the pieces together

H spreads a definite state into a superposition. Z flips the sign of the $|1\rangle$ amplitude. On their own, neither gate's effect is very dramatic once you already know the trick — so what happens if you slot Z into the exact middle of an H → H sandwich?

WORKED EXAMPLE — THREE GATES, STEP BY STEP

Step 1 (H): $|0\rangle \rightarrow (|0\rangle + |1\rangle)/\sqrt{2}$ — an equal superposition.

Step 2 (Z): $(|0\rangle + |1\rangle)/\sqrt{2} \rightarrow (|0\rangle - |1\rangle)/\sqrt{2}$ — the $|1\rangle$ amplitude flips sign; probabilities are still (0.5, 0.5).

Step 3 (H): apply H to $(|0\rangle - |1\rangle)/\sqrt{2}$. Using $H|0\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$ and $H|1\rangle = (|0\rangle - |1\rangle)/\sqrt{2}$:

$$H \cdot (|0\rangle - |1\rangle)/\sqrt{2} = (1/\sqrt{2}) \times [H|0\rangle - H|1\rangle] = (1/2) \times [(|0\rangle + |1\rangle) - (|0\rangle - |1\rangle)] = (1/2) \times [2|1\rangle] = |1\rangle.$$

Three gates turned a guaranteed $|0\rangle$ into a guaranteed $|1\rangle$ — with no measurement anywhere in between.

q0



Without the Z (just H → H) this circuit would return exactly to $|0\rangle$ instead — the only difference here is the Z in the middle.

WHY THIS IS SURPRISING

Z alone never changes a measurement — $|-1|^2 = 1$, same as $|+1|^2$. If you measured right after Z, you'd never notice it did anything. But hidden inside an $H \rightarrow _ \rightarrow H$ sandwich, that invisible sign flip becomes the entire reason the final answer changes from 0 to 1.

A FUN FACT

Multiply the three matrices together — $H \cdot Z \cdot H$ — and you get exactly the **X gate** (the quantum bit-flip / NOT gate). Sandwiching a phase gate between two Hadamards turns it into a completely different kind of gate. That's how central interference is: it doesn't just reveal phase, it can convert phase information into a flat-out different operation.

TAKEAWAY

$H \rightarrow Z \rightarrow H$ is not three independent steps — it's one combined operation (equivalent to X) that turns Z's invisible phase flip into a certain, visible outcome flip. The next section breaks down exactly why, path by path.

06**Follow the Interference**

Why the outcomes flip: path by path

"Interference" sounds abstract until you track it path by path. After the first H, the qubit has **two contributions** — one from the $|0\rangle$ branch, one from the $|1\rangle$ branch. The final H recombines both branches into *each* output. Whether they add up or cancel depends entirely on relative phase.

Splitting the final H by branch

Right before the last H, the state is $(|0\rangle - |1\rangle)/\sqrt{2}$ (from the $H \rightarrow Z \rightarrow H$ example above). Apply the final H to each term on its own, then add the results together for each output:

BRANCH	CONTRIBUTION TO $ 0\rangle$	CONTRIBUTION TO $ 1\rangle$
from the $ 0\rangle$ branch (weight $+1/\sqrt{2}$)	$+1/2$	$+1/2$
from the $ 1\rangle$ branch (weight $-1/\sqrt{2}$, after Z)	$-1/2$	$+1/2$
Total	$+1/2 - 1/2 = 0 \rightarrow$ destructive	$+1/2 + 1/2 = 1 \rightarrow$ constructive

- For the $|0\rangle$ outcome, the two branches arrive with **opposite sign** — they cancel exactly. This is **destructive interference**: zero chance of measuring 0.

- For the $|1\rangle$ outcome, the two branches arrive with the **same sign** — they reinforce exactly. This is **constructive interference**: certainty of measuring 1.
- Without the Z gate, both branches would agree on sign for *both* outputs' $|0\rangle$ contribution instead, so $|0\rangle$ would reinforce and $|1\rangle$ would cancel — the opposite pattern.

q0

 $(|0\rangle - |1\rangle)/\sqrt{2}$

H

recombined into $|1\rangle$ *splits into 2 paths**The last H is where the two branches meet and either cancel or reinforce, depending on their relative sign.***THE ONE-LINE RULE**

Every output is a sum over every path that can reach it. Same sign on both paths → amplitudes add → probability goes up. Opposite sign → amplitudes cancel → probability goes to zero. Relative phase is simply the thing that decides which paths agree and which disagree.

TAKEAWAY

Interference is bookkeeping over paths, not magic: add up every route to an outcome, and a single sign flip earlier in the circuit (like Z) is enough to turn a “definitely 0” circuit into a “definitely 1” one.